

Deep Learning: INTRODUCTION

Aprendizagem Profunda
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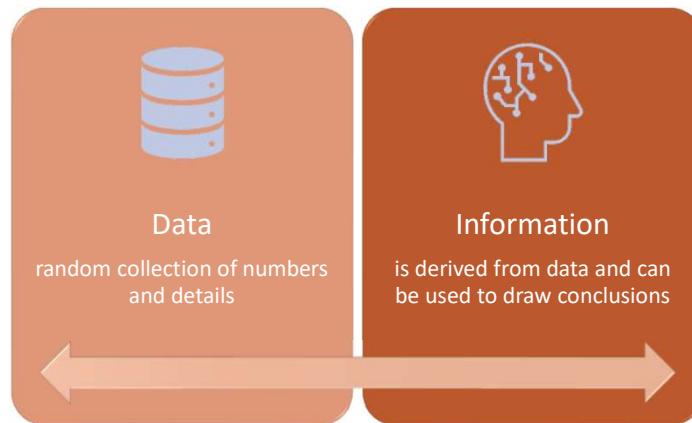
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From Data to Big Data

- The availability of **massive amounts of good data** and **fast computation** changes everything...
- From bytes to yottabytes: the data revolution
- Data-based decision making has emerged and evolved significantly
- New data is complex and unstructured!
 - Big Data
 - Data Science/Data Engineering/Data Analytics
 - Deep Learning

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Data is everywhere...



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What is Big Data?

... so much information in so many different forms ...

today ... significantly more data exists, and it varies in type and timeliness.

Need to managing data differently!

Data management has to include technology advances in hardware, storage, networking, and computing models (such as virtualization and cloud computing).

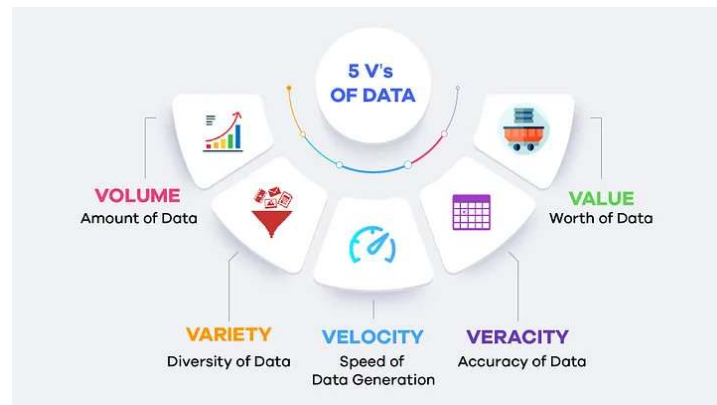
The **convergence of emerging technologies** and **reduction in costs** for have transformed the data landscape and made new opportunities possible.

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Data Science, Big Data, and Data Analytics

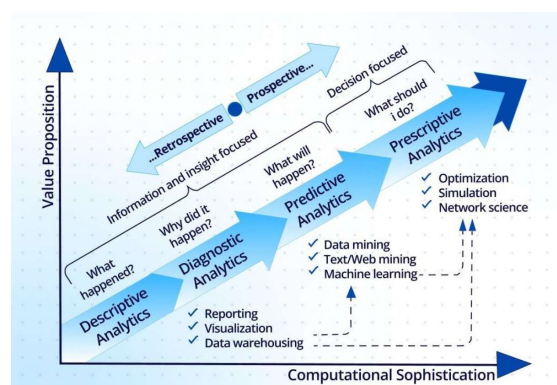
The characteristics of Big Data are often referred as the three V's:

- **Volume**—How much data is there?
- **Variety**—How diverse are different types of data?
- **Velocity**—At what speed is new data generated?



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Introduction to Machine Learning

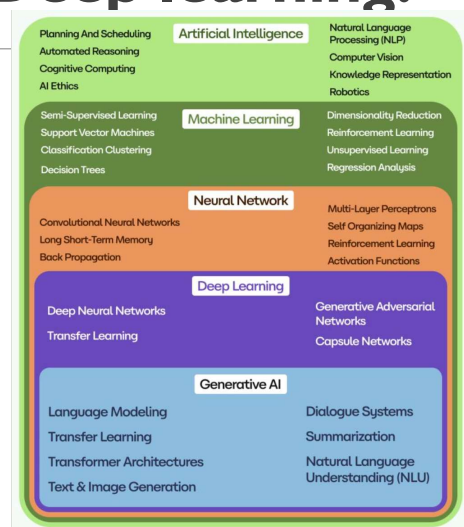
- Artificial intelligence
- Machine learning
- Deep learning
- Big Data
- Data Science/Data Engineering

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Machine learning vs Deep learning:

- **Artificial Intelligence** focuses on making machines intelligent.
- **Neural Networks** provide a structure inspired by the human brain.
- **Deep Learning** adds multiple layers to understand complex patterns.
- **Generative AI** takes it further by enabling models to create new content such as text, images and ideas.



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Introduction to Machine Learning



Artificial intelligence (AI) is considered as a subfield of computer science focusing on solving tasks that humans are good at (for example, natural language, image recognition) - mimic human intelligence.



Machine learning (ML), emerged as a subfield of AI – concerned with the development of algorithms so that computers can automatically learn (predictive) models from data.



Deep learning (DL), a subfield of machine learning, referring to a particular subset of models that are particularly good at certain tasks such as image recognition and natural language processing.



Big Data is high-volume, and high-velocity or high-variety information assets that demand cost-effective, innovative forms of information processing that enable enhanced insight, decision making, and process automation. data analysis, and much more.

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Traditional Machine Learning / LLMs

TRADITIONAL MACHINE LEARNING

Traditional ML **is task-first**:

- Define the problem, collect targeted data, design features, then train a model.
- Learns patterns from structured data solve one specific task.
- Needs a lot of human effort to choose features and guide learning.
- Learns from **labeled examples** and **fixed features**, and it excels with structured data like tables and numbers
- Train a model to do one task well like predicting prices or detecting fraud.
- Is the broader field that covers many **model types** and **techniques**



LLM'S

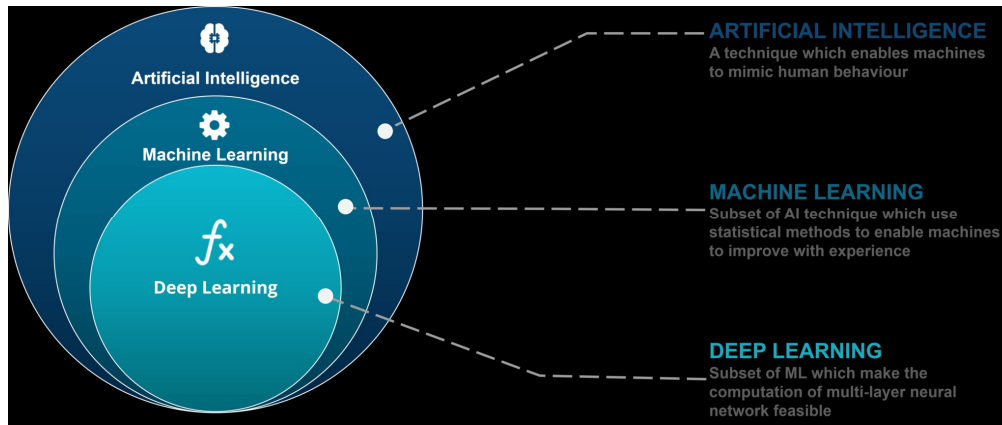
LLMs are **knowledge-first**:

- Build broad Knowledge by pretraining on massive data, then adapt with prompts or fine-tuning
- Learn by Reading massive amounts of text and predicting the next token, thriving on unstructured data like languages and documents
- Learn features automatically, so you focus more on what you want, not how to extract it.
- An LLM is a massive model trained on huge amounts of text to understand and generate language.
- Instead of one task, it learns general knowledge and adapts to many tasks, like answering questions or writing code.
- Sit on top of ML ideas, just scaled up and specialized for **language** and **reasoning**.



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Machine learning vs Deep learning:

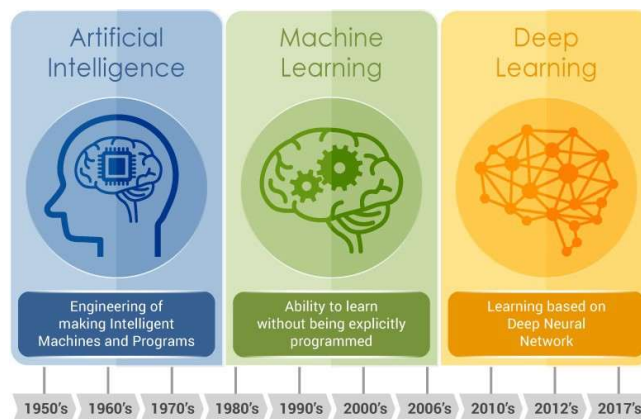


Fonte: <https://technical-beast.blogspot.com/2020/05/Artificial-Intelligence.html>

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Machine learning vs Deep learning:



Fonte: <https://medium.com/@mail2princeyadav/machine-learning-vs-deep-learning-b5c5a4fc5c>

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What is Machine Learning? An Overview

"Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed." - Arthur L. Samuel, AI pioneer, 1959

"The field of machine learning is concerned with the question of how to construct computer programs that automatically improve with experience". — Tom Mitchell, Professor Machine Learning at Carnegie Mellon University and author of the popular "Machine Learning" textbook

Machine learning is the hot new thing. — John L. Hennessy, President of Stanford (2000–2016)

A breakthrough in machine learning would be worth ten Microsofts. — Bill Gates, Microsoft Co-Founder

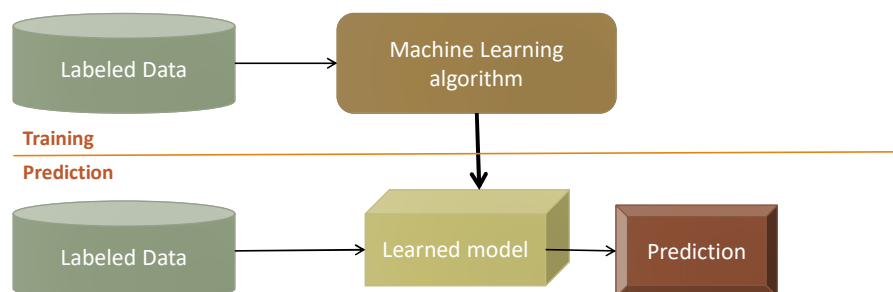


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Machine Learning Basics

Machine learning is a field of computer science that gives computers the ability to **learn without being explicitly programmed**

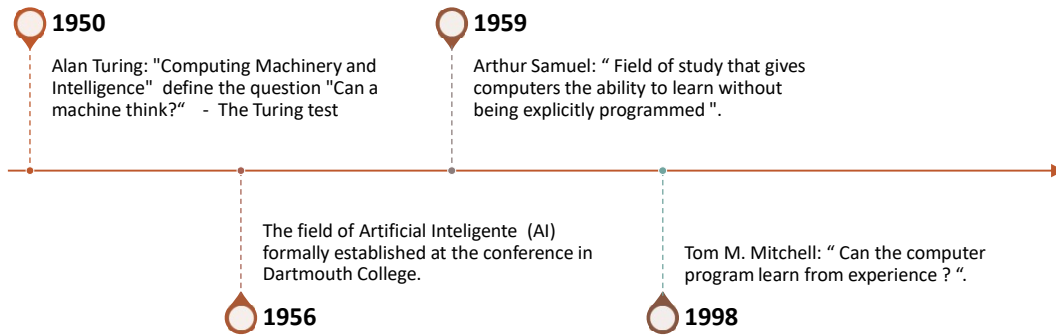


Methods that can learn from and make predictions on data

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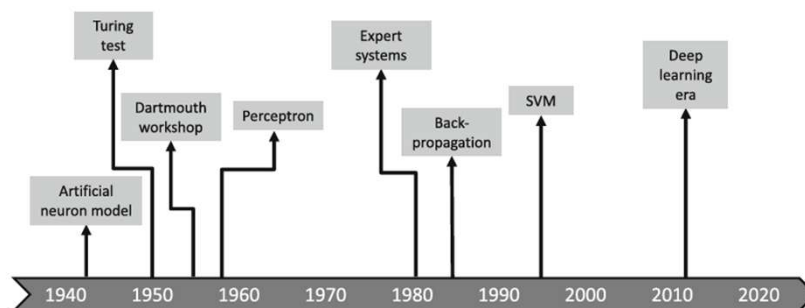
Machine Learning Historic Perspective



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Artificial Intelligence Historic Perspective



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Why Machine Learning (ML)?

The real world is complex – difficult to hand-craft solutions.



ML is the preferred framework for applications in many fields:

Computer Vision

Natural Language Processing,
Speech Recognition

Robotics

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Components of Machine Learning

Datasets are required for training machine learning programs to draw patterns and correlations. Such datasets include images, numbers, texts and other forms of data.

Features, also known as variables, highlight key pieces of data the program should focus on. Selecting the right features is crucial in training the software to make the right decisions.

Algorithms are methods of analysing data. While using different algorithms for the same task may provide identical solutions, the speed and accuracy of obtaining the results may differ.

- The precision of the results rely on the quality of data
- Collecting accurate data will aid in achieving the desired outcomes.

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Data Science, Big Data, and Data Analytics

Big Data is high-volume, and high-velocity or high-variety information assets that demand cost-effective, innovative forms of information processing that enable enhanced insight, decision making, and process automation, data analysis, and much more.

Data Engineering is concerned with the design and construction of systems for collecting, storing, and processing data. It forms the basis on which data analysis and machine learning models are built.

Data Science is an interdisciplinary field that uses scientific methods, processes, algorithms, and systems to extract knowledge and insights from structured and unstructured data.

Data Analytics is the process of examining raw data to reach certain conclusions.

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Comparison of Machine Learning and Deep Learning

Hardware Dependencies

- DL requires high-end machines because while doing features extractions and classification at different part of hidden layers requires lot of large matrix multiplication.
- Traditional ML algorithms, can work on low-end machines.

Data Dependencies - Performance of both techniques differ as the scale of data increases.

- When data is less DL doesn't perform very well because of the huge number of parameters needed to be tuned by a learning algorithm.
- DL starts with some random parameters and then some gradient based optimization algorithm is used to converge the network to an optimum solution, which might not be global optimum. This whole process requires a lot of data.
- DNN have many false positive initially and slightly improves with every learning iteration.
- On the other hand ML algorithm have their handcrafted rules which works in less amount of data.

Feature Extraction

- In ML, most of the applied features need to be identified by an experts and then hard-coded as per the domain and data type. Eg. features can be pixels values, textures, shape, position and orientation.
- The performance of most of the ML algorithm depends on how accurately the features are identified and extracted.
- DL algorithms try to learn high-level features from data. This is very distinctive part of DL and a major step ahead of traditional ML.
- DL reduces the task of developing a new feature extractor of every problem. Eg. CNN (Convolutional Neural Network) will try to learn low-level features such as edge and lines at early layers and then high level features in next hidden layers.

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Deep Learning

Deep learning techniques are so powerful because they learn the best way to represent the problem while learning how to solve the problem.

This is called **representation learning**.

Representation learning is perhaps the biggest differentiation between deep learning models and classical machine learning algorithm.

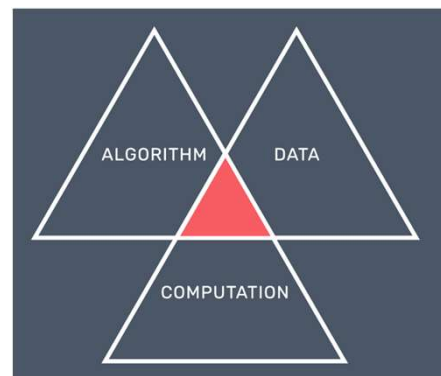
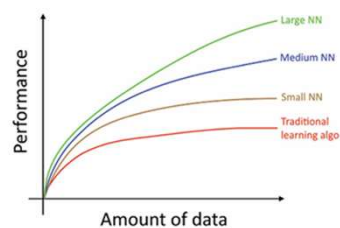
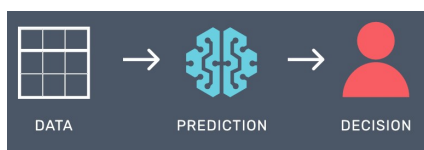
It is the power of representation learning that is spurring such great creativity in the way the techniques are being used. For example:

- **Deep learning** models are being used for very difficult problems and making progress, like colorizing image and videos based on the context in the scene.
- **Deep learning** models are being used in bold new ways, such as cutting the head off a network trained on one problem and tuning it for a completely different problem, and getting impressive results.
- Combinations of deep learning models are being used to both identify objects in photographs and then generate textual descriptions of those objects, a complex multi-media problem that was previously thought to require large artificial intelligence systems.

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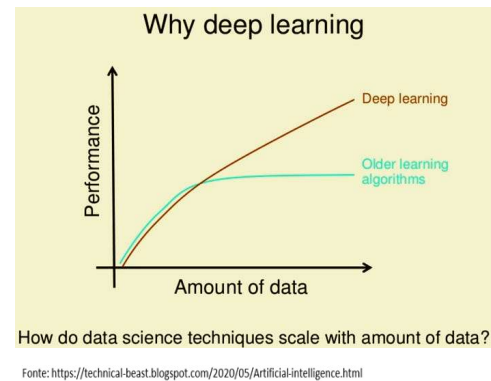
Big Data – Deep Learning



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When use Deep Learning ?

- Deep Learning requires high-end machines contrary to traditional ML algorithms. GPU has become an integral part to execute Deep Learning algorithm.
- DL outperform ML if the data size is large. With small data size, traditional ML algorithms are preferable.
- DL techniques need to have high end infrastructure to train in reasonable time. Requires high-end machines contrary to traditional ML algorithms. GPU has become an part to execute Deep Learning algorithm.
- When there is lack of domain understanding for feature introspection.
- To complex problems such as image classification, natural language processing, and speech recognition.



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Machine Learning vs Deep Learning

Machine Learning	Deep Learning
Machine Learning is a superset of Deep Learning	Deep Learning is a subset of Machine Learning
The data represented in Machine Learning is quite different compared to Deep Learning as it uses structured data	The data representation used in Deep Learning is quite different as it uses neural networks(ANN).
Machine Learning is an evolution of AI.	Deep Learning is an evolution of Machine Learning. Basically, it is how deep is the machine learning.
Machine learning consists of thousands of data points.	Big Data: Millions of data points.
Outputs: Numerical Value, like classification of the score.	Anything from numerical values to free-form elements, such as free text and sound.
Uses various types of automated algorithms that turn to model functions and predict future action from data.	Uses a neural network that passes data through processing layers to, interpret data features and relations.
Algorithms are detected by data analysts to examine specific variables in data sets.	Algorithms are largely self-depicted on data analysis once they're put into production.
Machine learning algorithms for complex tasks, but they can also be more difficult to train and may require more computational resources.	Deep learning algorithms are more accurate than machine learning algorithms.

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Machine Learning vs Deep Learning

Machine Learning	Deep Learning
Machine Learning is highly used to stay in the competition and learn new things.	Deep Learning solves complex machine-learning issues.
Training can be performed using the CPU (Central Processing Unit).	A dedicated GPU (Graphics Processing Unit) is required for training.
More human intervention is involved in getting results.	Although more difficult to set up, deep learning requires less intervention once it is running.
Machine learning systems can be swiftly set up and run, but their effectiveness may be constrained.	Although they require additional setup time, deep learning algorithms can produce results immediately (although the quality is likely to improve over time as more data becomes available).
Its model takes less time in training due to its small size.	A huge amount of time is taken because of very big data points.
Humans explicitly do feature engineering.	Feature engineering is not needed because important features are automatically detected by neural networks.
Machine learning applications are simpler compared to deep learning and can be executed on standard computers.	Deep learning systems utilize much more powerful hardware and resources.

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Machine Learning vs Deep Learning

Machine Learning	Deep Learning
The results of an ML model are easy to explain.	The results of deep learning are difficult to explain.
Machine learning models can be used to solve straightforward or a little bit challenging issues.	Deep learning models are appropriate for resolving challenging issues.
Banks, doctor's offices, and mailboxes all employ machine learning already.	Deep learning technology enables increasingly sophisticated and autonomous algorithms, such as self-driving automobiles or surgical robots.
Machine learning involves training algorithms to identify patterns and relationships in data.	Deep learning, on the other hand, uses complex neural networks with multiple layers to analyze more intricate patterns and relationships.
Machine learning algorithms can range from simple linear models to more complex models such as decision trees and random forests.	Deep learning algorithms, on the other hand, are based on artificial neural networks that consist of multiple layers and nodes.
Machine learning algorithms typically require less data than deep learning algorithms, but the quality of the data is more important.	Deep learning algorithms, on the other hand, require large amounts of data to train the neural networks but can learn and improve on their own as they process more data.
Machine learning is used for a wide range of applications, such regression, classification and clustering.	Deep learning, on the other hand, is mostly used for complex tasks such as image and speech recognition, natural language processing, and autonomous systems.

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What deep learning has achieved so far?

- Near-human-level image classification
- Near-human-level speech transcription
- Near-human-level handwriting transcription
- Dramatically improved machine translation
- Dramatically improved text-to-speech conversion
- Digital assistants such as Google Assistant and Amazon Alexa
- Near-human-level autonomous driving
- Improved ad targeting, as used by Google, Baidu, or Bing
- Improved search results on the web
- Ability to answer natural language questions
- Superhuman Go playing

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What is Deep Learning?

It is a specific area of machine learning

- Supervised learning
- Unsupervised learning
- Reinforcement learning

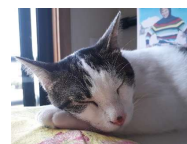
Idea (supervised learning): learn how to make decisions, perform a task, from examples



dog



cat

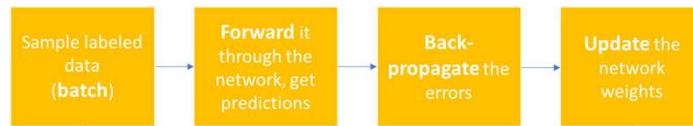


dog or cat?

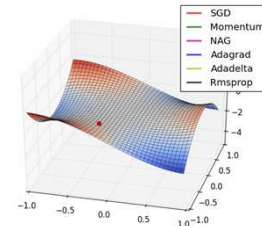
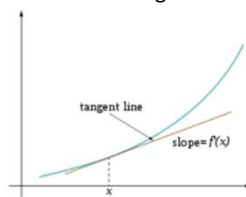
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Optimization of Deep Neural Networks



- Optimize (min. or max.) **objective/cost function $J(\theta)$**
- Generate **error signal** that measures difference between predictions and target values



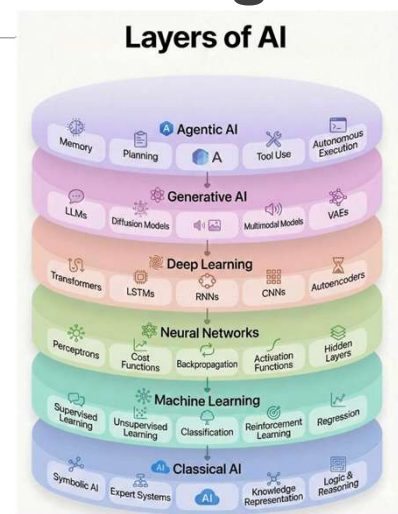
- <https://medium.com/@ramrajchandradevan/the-evolution-of-gradient-descent-optimization-algorithm-4106a6702d39>
- Use error signal to change the **weights** and get more accurate predictions
 - Subtracting a fraction of the **gradient** moves you towards the **(local) minimum of the cost function**

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Machine learning vs Deep learning:

- Artificial Intelligence** focuses on making machines intelligent.
- Machine learning (ML)**, emerged as a subfield of AI – concerned with the development of algorithms so that computers can automatically learn (predictive) models from data.
- Neural Networks** provide a structure inspired by the human brain.
- Deep Learning** adds multiple layers to understand complex patterns.
- Generative AI** takes it further by enabling models to create new content such as text, images and ideas.
- Agentic AI** - or autonomous AI, is a type of artificial intelligence that runs independently to design, execute, and optimize workflows



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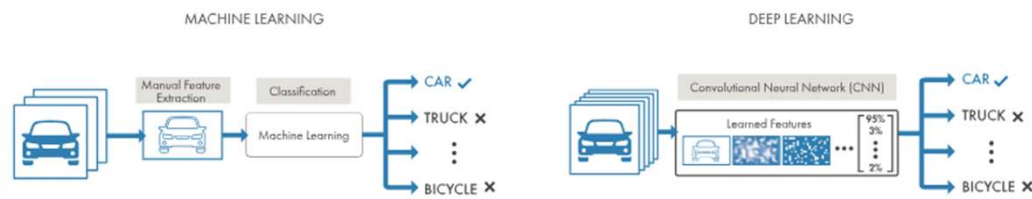
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What is Deep Learning (DL) ?

A machine learning subfield of learning **representations** of data. Exceptional effective at **learning patterns**.

Deep learning algorithms attempt to learn (multiple levels of) representation by using a **hierarchy of multiple layers**

If you provide the system **tons of information**, it begins to understand it and respond in useful ways



<https://www.mathworks.com/discovery/deep-learning.html>

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Why Deep Learning ?

- Deep Learning describes algorithms that analyze data with a logical structure similar to how a human would draw conclusions.
- Through supervised and unsupervised learning.
- For that, Deep Learning applications use a layered structure of algorithms called an artificial neural network (ANN).
- The design of an ANN is inspired by the biological neural network of the human brain, leading to a process of learning that's far more capable than that of standard machine learning models.
- The choice of the architecture and type of an ANN depends on the type of application and the type of data.

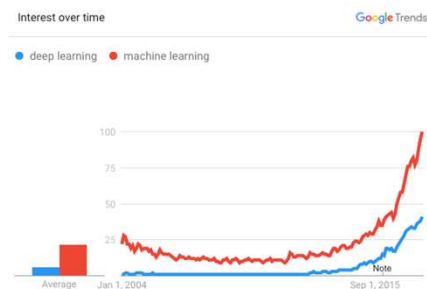
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Why is DL useful?

- Manually designed features are often **over-specified, incomplete** and take a **long time to design** and validate
- Learned Features are **easy to adapt, fast** to learn
- Deep learning provides a very **flexible**, (almost?) **universal**, learnable framework for representing world, visual and linguistic information.
- Can learn both unsupervised and supervised
- Effective **end-to-end** joint system learning
- Utilize large amounts of training data

In ~2010 DL started outperforming other ML techniques
first in speech and vision, then NLP



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Deep learning and neural networks are related concepts, but they are not exactly the same



Neural networks are a type of machine learning algorithm that are inspired by the structure and function of the human brain. They consist of a large number of interconnected artificial neurons, which are organized into layers. Neural networks can be used for a wide range of tasks, such as image recognition, speech recognition, and natural language processing.



Deep learning is a specific type of neural network that is characterized by its depth, which means that it has many layers, usually more than 10 layers. Deep learning networks are able to extract features from the input data and to learn complex relationships between the features and the output labels. They are particularly well-suited for tasks that involve large and complex datasets, such as image and speech recognition, natural language processing, and computer vision.



Deep learning is a subset of neural network, deep learning is the umbrella term that refers to the use of neural networks with multiple hidden layers, these layers are able to learn hierarchical representations of the data, which makes them more powerful than traditional neural networks.



Deep learning has shown to be very effective in several fields, like computer vision, natural language processing, speech recognition, and bioinformatics, because of its ability to learn from large and complex datasets, and its ability to automatically extract features from the input data.

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Deep learning is a specific type of neural network that is characterized by its depth, which means that it has many layers, usually more than 10 layers. Deep learning has become important in recent years due to its ability to learn from large and complex datasets, and its ability to automatically extract features from the input data.

Why Deep Learning is important?

Here are some reasons why deep learning is important:

- **Improved performance:** Deep learning networks have been shown to achieve state-of-the-art performance on a wide range of tasks, such as image classification, speech recognition, and natural language processing. This is because deep learning networks are able to learn complex relationships between the input data and the output labels, and they can handle large and complex datasets.
- **Automated feature extraction:** One of the main advantages of deep learning is that it can automatically extract features from the input data, which means that the network can learn to recognize patterns and features that are relevant to the task, without the need for manual feature engineering.
- **Handling unstructured data:** Deep learning can handle unstructured data, such as images, videos, and text, which makes it a powerful tool for tasks such as image recognition, speech recognition, and natural language processing.
- **Handling noisy and incomplete data:** Deep learning can handle noisy and incomplete data, which makes it a powerful tool for tasks such as anomaly detection, fault diagnosis, and sensor **data analysis**.
- **Handling complex non-linear relationships:** Deep learning can handle complex non-linear relationships between input and output data, which makes it a powerful tool for tasks such as time series prediction, drug discovery, and finance.

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